

MBA Fastrack 2025 (CAT + OMETs)

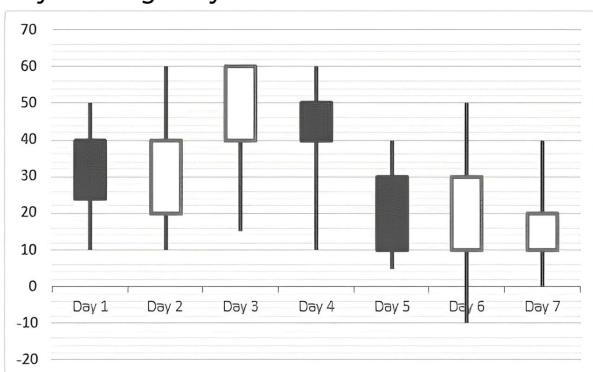
Data Interpretation & Logical Reasoning

DPP: 4

Miscellaneous Charts - 2

Directions (1-5) Read the following passage and answer the given questions.

Ankit, a stock broker, who does “One-day trading”, which means that he buys stocks in the morning and sells them during the day. He invests in 7 different stocks on 7 different days of a week. Units of stocks are called shares. The shares of every stock purchased by Ankit from Day 1 through Day 7 are consecutive integral multiples of 10 in the given order. The candlestick chart depicts the prices of these 7 stocks on the day that each one was purchased. The top and bottom ends of the line respectively indicate the maximum and minimum prices of the stock at any time on that day. The horizontal edges of the rectangle correspond to the stock's opening and closing prices. If the rectangle is white, then the opening price is lower than the closing price, but if the rectangle is shaded, then the closing price is lower than the opening price. Ankit buys the shares of the stocks at the opening time, 9:00 AM and sells the shares of each stock at the closing time, 6:00 PM. Further known information is that Ankit buys shares of each stock in a distinct consecutive quantity less than 100 in the integral multiples of 10 in the given order from Day 1 through day 7.



Note: Person has to have enough money in the wallet because the Share price can be negative and if share sold at negative price, money will be deducted from the wallet accordingly.

- Q1** If the profit earned by Ankit at the end of 2 out of 7 days during the week was Rs.800 each, then what was the total number of shares of each stock purchased by him during the week?
 (A) 350 (B) 280
 (C) 380 (D) 420
- Q2** If Shweta, who is a friend of Ankit, invests in a similar manner as Ankit, but buys a lesser number of shares of each stock than Ankit, then what will be the maximum total profit (in Rs.) earned by Shweta and Ankit together at any time on the second day of the week?
 (A) 3500 (B) 1400
 (C) 2800 (D) 2900
- Q3** If Ankit earned a profit of Rs. 1,600 on a particular day of the week, then what was the total returns (in Rs) earned by him at the end of each day during the whole week?
 (A) 1460 (B) 1620
 (C) 1140 (D) 1820
- Q4** If Ankit earned a total profit of Rs. 1,580 in the week, then what was the total minimum value (in Rs.) of share of each stock bought by Ankit at any time of the day in the week?
 (A) 1900 (B) 1200
 (C) 1120 (D) 1400
- Q5** If Ankit had a loss of Rs. 400 on a day in a week, then what was the total loss or profit

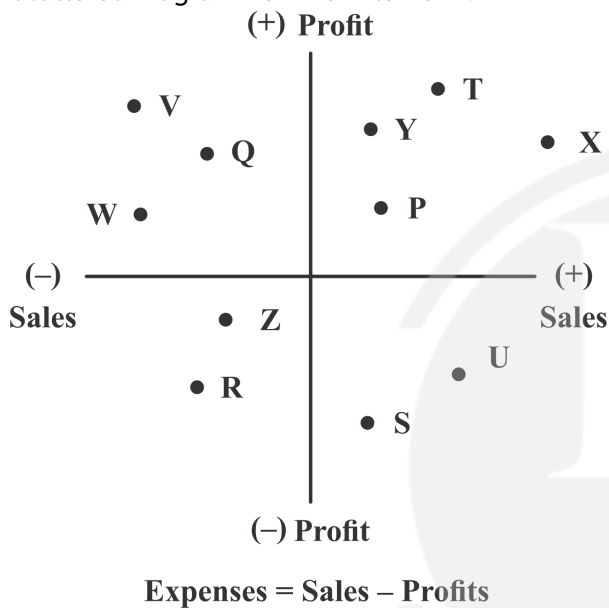


(in Rs.) earned by Ankit at the end of each day in that week?

- (A) 1820 (B) 1340
(C) 1460 (D) 1580

Directions (6-10) Read the following passage and answer the given questions.

The sale and profit of eleven companies P, Q, R, S, T, U, V, W, X, Y and Z are observed by the GST inspector and the change in sales and profit of these companies is represented by the following Scattered Diagram from 2021 to 2022.



Q6 Find the number of companies for which, did the expenses increase definitely from 2021 to 2022?

- (A) 5
(B) 4
(C) 2
(D) Cannot be determined

Q7 Find the number of companies for which, did the expenses decrease definitely from 2021 to 2022?

- (A) 3 (B) 5
(C) 4 (D) 6

Q8 What could be the expense of company X in 2022, if it was ₹1000 Crores in 2021?

- (A) ₹ 990 Crores (B) ₹ 1100 Crores
(C) ₹ 1050 Crores (D) Either B or C

Q9 What could be the expense of company Q in 2022, if it was ₹100 Crores in year 2021?

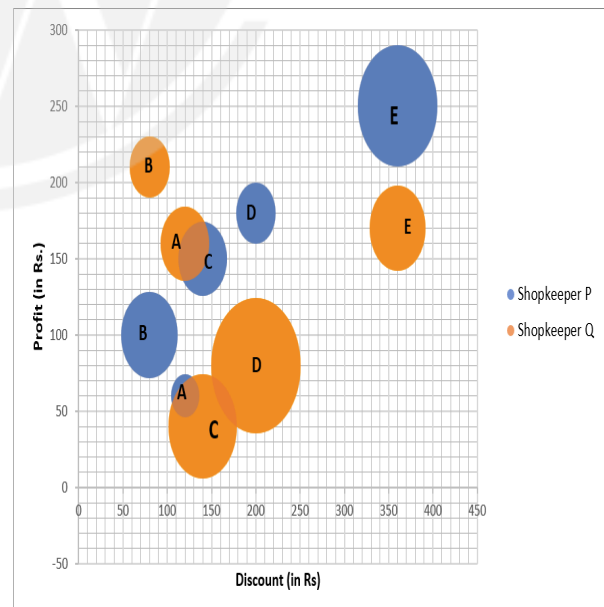
- (A) ₹ 110 Crores (B) ₹ 85 Crores
(C) ₹ 125 Crores (D) ₹ 100 Crores

Q10 The expenditure of all the companies were the same in the year 2021. What can be said about their expenditure in 2022?

- (A) Expenditure of S = Expenditure of Q
(B) Expenditure of S > Expenditure of Q
(C) Expenditure of S < Expenditure of Q
(D) None of these

Directions (11-15) Read the following passage and answer the given questions.

Two shopkeepers P and Q bought 5 different types of items at different prices. Following bubble chart represents the data regarding discounts given by these shopkeepers on the marked price of different items and profit earned by them on respective items. X-Y coordinates at the center of each bubble represents the discount given (in rupees) and profit earned (in rupees) on each item. Numerical value of area of bubble represents the marked price of each item. (Take $\pi = 3.14$)



The following table represents the ratio of the radius of the bubble for each item for shopkeeper P and Q and their difference -



Item	Ratio of radius (Shopkeeper P : Shopkeeper Q)	Difference between radius of bubble for shopkeeper P and Q
A	1 : 3	50
B	2 : 1	50
C	1 : 2	75
D	1 : 5	200
E	2 : 1	100

Q11 What is the difference between the CP of Product C for shopkeeper P and Q?

- (A) 54545 (B) 52335
(C) 53097.5 (D) 53000

Q12 Cost price of all given items taken together for shopkeeper P is approximately what percent more/less than the cost price of all given items taken together for shopkeeper Q?

- (A) 45% (B) 44%
(C) 43% (D) 42%

Q13 If the radius of bubble D for shopkeeper P and Q are decreased by 15% and 40% respectively, then what will be the approximate difference between new selling price (Rs.) of item D sold by shopkeeper P and Q?

- (A) 63867 (B) 64978
(C) 65089 (D) 66180

Q14 If shopkeeper P and Q sells item D at Rs.4500 and Rs.5200 below the original CP respectively, then the average of new selling price of item D for shopkeeper P and Q together is how much (in INR) ?

- (A) 97680 (B) 96870
(C) 97860 (D) 98760

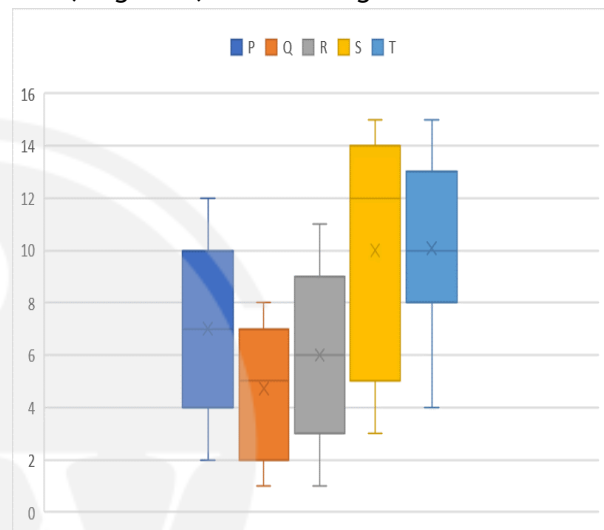
Q15 What is the difference between the cost prices of item B for the shopkeeper P and Q?

- (A) 21640 (B) 22650
(C) 23660 (D) 24670

Directions (16-20) Read the following passage and answer the given questions.

During an experiment five power generators P, Q, R, S, and T are used and the experiment is conducted for a limited time. The graph given below shows the maximum and minimum power (MW) of power generated by those generators and it also shows the power generated by those generators when the experiment starts and when it ends.

Note. All the power measures are calculated in MW (megawatt) and are integers.



The ceiling (top) of the rectangle shows the power generated when the experiment starts, and floor (bottom) of the rectangle shows the power generated when the experiment ends.

The ceiling (top) of the line shows the maximum power generated by the generators while the floor (bottom) of the line shows the minimum power generated by the generators during the experiment.

There is also some loss of power generated by the generator: Loss in power when the generator generates more than 5 MW is 20% and otherwise, it is 25%.

Note: 1 MW = 1000 KW and 1 KW = 1000 W

Q16 If the generator P when generating minimum power can run 'x' machine that uses 12 KW power and generator Q at the end of the experiment can run 'y' machines



that uses 10 KW power, then what is the ratio of 'x' to 'y'?

- (A) 3: 8 (B) 5 : 6
(C) 4 : 9 (D) 2 : 5

Q17 Generator R when generating maximum power can run 'x' machines that uses 88 KW power when there is no problem in the machines. If because of some problem, those machines started consuming 12 KW more power, then by what percent number of machines are decreased to use power generated by R at that time?

- (A) 10% (B) 12%
(C) 14% (D) 15%

Q18 Type A machines use 12 KW power and type B machines use 13 KW power. If the ratio of total type A to type B machines that can be run from the generator S when the experiment ends is 1: 1, then what is the total number of machines used?

- (A) 392 (B) 300
(C) 312 (D) 320

Q19 If the generator S when generating minimum power can run 'x' machine that uses 15 KW power and generator T when generating maximum power can run 'y' machines that uses 24 KW power, then what is the ratio of 'x' to 'y'?

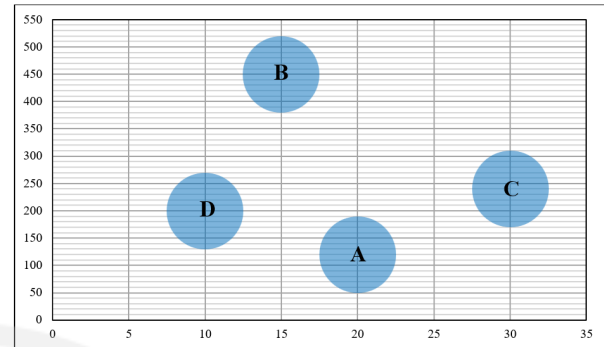
- (A) 3 : 5 (B) 3 : 10
(C) 2 : 7 (D) 2 : 21

Q20 Generator T when generating maximum power can run 'x' machines that use 60 KW power when there is no problem in the machines. If because of some problem, those machines started consuming 40 KW more power, then by what percent number of machines are decreased to use power generated by T at that time?

- (A) 50% (B) 45%
(C) 40% (D) 35%

Directions (21-25) Read the following passage and answer the given questions.

The given scatter graph shows the total capacity of four dams (in hundred litres) (y – axis) and amount of electricity (in units) which can be generated (x – axis) from 100 litres of water in the respective dams.



Q21 Dam 'A' is filled up to 45% of its capacity. If only 50% of water present in dam 'A' was used to produce electricity then find the amount of electricity which can be generated from dam 'A'.

- (A) 320 units (B) 480 units
(C) 540 units (D) 640 units

Q22 Find the ratio of the amount of electricity which can be generated from dams 'B' and 'D' when they are full.

- (A) 20 : 7 (B) 21 : 4
(C) 12 : 5 (D) 27 : 8

Q23 Dam 'C' is emptied by 6000 litres. The amount of electricity which can be generated from water available in dam 'C' has to be sold for Rs. 5/unit. Find the total amount which can be earned from the electricity generated from dam 'C'.

- (A) Rs 27000 (B) Rs 32000
(C) Rs 24000 (D) Rs 30000

Q24 Find the average capacities of given four dams.

- (A) 20450 litres (B) 26250 litres
(C) 24350 litres (D) None of these

Q25



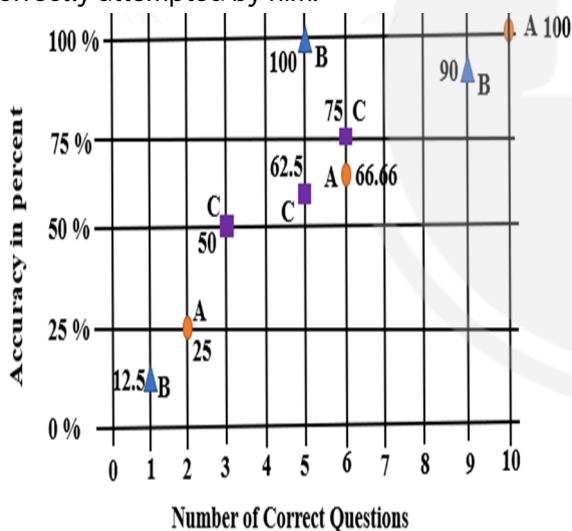
Find the difference between the amount of electricity which can be produced when dams 'B' and dam 'D' are full to their capacity.

- (A) 3840 units (B) 4750 units
(C) 4080 units (D) 4250 units

Directions (26-30) Read the following passage and answer the given questions.

Three groups of students, A, B and C each comprising of 3 members, participate in a Maths competition. Each member of all the groups gets a set of 10 objective questions to solve. A group's performance is evaluated by calculating the ratio of total number of questions answered correctly to the total number of questions attempted by all members combined. The group with the best performance wins.

Following graph shows the percent accuracy of a member against the number of questions correctly attempted by him.



Q26 The winner of the Maths Competition is-

- (A) Group C
(B) Group A
(C) Group B
(D) Cannot be determined

Q27 Maximum number of attempts made by-

- (A) Group A
(B) Group B
(C) Group C
(D) Cannot be determined

Q28 If exactly one member from Group A and exactly one member from Group C had swapped their groups, then the performance of Group C would have been higher than that of Group A. What was the percent accuracy of the member from Group C who moved to Group A to make this happen?

- (A) 75%
(B) 50%
(C) 62.5%
(D) More than one of the above

Q29 If '4' marks are to be awarded for every correct answer and there is a penalty of '1' mark for every incorrect answer, what is the total score of Group B?

Q30 At least how many more questions (out of attempted questions only) should be answered correctly by the worst performer of Group B so that Group B would have the winner?

- (A) 1 (B) 2
(C) 3 (D) 4



Answer Key

Q1 (A)
Q2 (C)
Q3 (D)
Q4 (B)
Q5 (B)
Q6 (B)
Q7 (D)
Q8 (D)
Q9 (B)
Q10 (B)
Q11 (C)
Q12 (C)
Q13 (B)
Q14 (B)
Q15 (C)

Q16 (B)
Q17 (B)
Q18 (B)
Q19 (B)
Q20 (C)
Q21 (C)
Q22 (D)
Q23 (A)
Q24 (D)
Q25 (B)
Q26 (B)
Q27 (A)
Q28 (D)
Q29 52
Q30 (A)



Hints & Solutions

Note: scan the QR code to watch video solution

Q1. Text Solution:

As per the candle-stick chart:

	Open	High	Low	Close	Difference	Multiply by 10
Day 1	40	50	10	24	-16	-160
Day 2	20	60	10	40	20	200
Day 3	40	60	15	60	20	200
Day 4	50	60	10	40	-10	-100
Day 5	30	40	5	10	-20	-200
Day 6	10	50	-10	30	20	200
Day 7	10	40	0	20	10	100

Let's assume the number of shares bought each day is:

Day 1: 10 shares

Day 2: 20 shares

Day 3: 30 shares

Day 4: 40 shares

Day 5: 50 shares

Day 6: 60 shares

Day 7: 70 shares

Now, let's recalculate the profit for the days with positive differences:

Day 2: $20 \times 20 = 400$ (Not Rs. 800)

Day 3: $20 \times 30 = 600$ (Not Rs. 800)

Day 6: $20 \times 60 = 1200$ (Not Rs. 800)

Day 7: $10 \times 70 = 700$ (Not Rs. 800)

Still, none of these days give a profit of Rs. 800.

This suggests that the number of shares bought each day might be different.

Let's try a different set of consecutive multiples of 10:

Day 1: 20 shares

Day 2: 30 shares

Day 3: 40 shares

Day 4: 50 shares

Day 5: 60 shares

Day 6: 70 shares

Day 7: 80 shares

Now, let's recalculate the profit for the days with positive differences:

Day 2: $20 \times 30 = 600$ (Not Rs. 800)

Day 3: $20 \times 40 = 800$ (Rs. 800)

Day 6: $20 \times 70 = 1400$ (Not Rs. 800)

Day 7: $10 \times 80 = 800$ (Rs. 800)

Now, we have two days (Day 3 and Day 7) where the profit is Rs. 800 each.

Now, let's sum up the number of shares bought each day:

Day 1: 20 shares

Day 2: 30 shares

Day 3: 40 shares

Day 4: 50 shares

Day 5: 60 shares

Day 6: 70 shares

Day 7: 80 shares

Total shares = $20 + 30 + 40 + 50 + 60 + 70 + 80 = 350$ shares

Video Solution:



Q2. Text Solution:

	Open	High	Low	Close	Difference	Multiply by 10
Day 1	40	50	10	24	-16	-160
Day 2	20	60	10	40	20	200



Day 3	40	60	15	60	20	200
Day 4	50	60	10	40	-10	-100
Day 5	30	40	5	10	-20	-200
Day 6	10	50	-10	30	20	200
Day 7	10	40	0	20	10	100

If Ankit buys 40 shares and Shweta buys 30 shares on Day 2, all the conditions of the problem are satisfied:

- The number of shares bought by Ankit and Shweta are consecutive multiples of 10 (30 and 40).
- Shweta buys fewer shares than Ankit for each stock ($30 < 40$).
- The number of shares bought each day is distinct and less than 100.

From the table, the data for Day 2 is:

Open: 20

High: 60

Low: 10

Close: 40

The maximum profit at any point in time on Day 2 is calculated using the High and Open prices:

Maximum Profit = (High – Open) × Number of Shares

Ankit's Maximum Profit on Day 2:

Profit Ankit = $(60 - 20) \times 40$

= 40×40

= 1600 Rs.

Shweta's Maximum Profit on Day 2:

Profit Shweta = $(60 - 20) \times 30$

= 40×30

= 1200 Rs.

Total Maximum Profit = Profit Ankit + Profit Shweta

= $1600 + 1200$

= 2800 Rs.

The maximum total profit earned by Shweta and

Ankit together at any point in time on Day 2 is 2800 Rs.

Video Solution:



Q3. Text Solution:

	Open	High	Low	Close	Difference	Multiply by 10
Day 1	40	50	10	24	-16	-160
Day 2	20	60	10	40	20	200
Day 3	40	60	15	60	20	200
Day 4	50	60	10	40	-10	-100
Day 5	30	40	5	10	-20	-200
Day 6	10	50	-10	30	20	200
Day 7	10	40	0	20	10	100

If Ankit buys the following number of shares each day:

Day 1: 30 shares

Day 2: 40 shares

Day 3: 50 shares

Day 4: 60 shares

Day 5: 70 shares

Day 6: 80 shares

Day 7: 90 shares

Now, let's recalculate the profit for each day:

Day 1:

Open: 40

Close: 24

Difference: -16

Profit: $-16 \times 30 = -480$ (Loss)



Day 2:

Open: 20

Close: 40

Difference: $40 - 20 = 20$ Profit: $20 \times 40 = 800$ **Day 3:**

Open: 40

Close: 60

Difference: $60 - 40 = 20$ Profit: $20 \times 50 = 1000$ **Day 4:**

Open: 50

Close: 40

Difference: $40 - 50 = -10$ Profit: $-10 \times 60 = -600$ (Loss)**Day 5:**

Open: 30

Close: 10

Difference: $10 - 30 = -20$ Profit: $-20 \times 70 = -1400$ (Loss)**Day 6:**

Open: 10

Close: 30

Difference: $30 - 10 = 20$ Profit: $20 \times 80 = 1600$ **Day 7:**

Open: 10

Close: 20

Difference: $20 - 10 = 10$ Profit: $10 \times 90 = 900$

Now, we see that on Day 6, Ankit earns a profit of Rs. 1,600.

Now, let's sum up the profits for all 7 days:

Total Returns: $-480 + 800 + 1000 - 600 - 1400 + 1600 + 900 = 1820$ Rs.

Video Solution:**Q4. Text Solution:**

	Open	High	Low	Close	Difference	Multiply by 10
Day 1	40	50	10	24	-16	-160
Day 2	20	60	10	40	20	200
Day 3	40	60	15	60	20	200
Day 4	50	60	10	40	-10	-100
Day 5	30	40	5	10	-20	-200
Day 6	10	50	-10	30	20	200
Day 7	10	40	0	20	10	100

Ankit earned a total profit of Rs. 1,580 in the week.

Hence, Ankit must buy the following number of shares each day:

Day 1: 20 shares

Day 2: 30 shares

Day 3: 40 shares

Day 4: 50 shares

Day 5: 60 shares

Day 6: 70 shares

Day 7: 80 shares

It is given that the share price can be negative, and selling at a negative price deducts money from the wallet, we need to account for this in our calculations.

Minimum Value = Low $\times$ Number of Shares

Day 1:

Low: 10

Number of Shares: 20

Minimum Value: $10 \times 20 = 200$ **Day 2:**

Low: 10

Number of Shares: 30

Minimum Value: $10 \times 30 = 300$ **Day 3:**

Low: 15

Number of Shares: 40



Minimum Value: $15 \times 40 = 600$

Day 4:

Low: 10

Number of Shares: 50

Minimum Value: $10 \times 50 = 500$

Day 5:

Low: 5

Number of Shares: 60

Minimum Value: $5 \times 60 = 300$

Day 6:

Low: -10

Number of Shares: 70

Minimum Value: $-10 \times 70 = -700$ (Money will be deducted from the wallet.)

Day 7:

Low: 0

Number of Shares: 80

Minimum Value: $0 \times 80 = 0$

Sum of the minimum values for all 7 days: $200 + 300 + 600 + 500 + 300 - 700 + 0 = 1200$ Rs.

Video Solution:**Q5. Text Solution:**

	Open	High	Low	Close	Difference	Multiply by 10
Day 1	40	50	10	24	-16	-160
Day 2	20	60	10	40	20	200
Day 3	40	60	15	60	20	200
Day 4	50	60	10	40	-10	-100
Day 5	30	40	5	10	-20	-200
Day 6	10	50	-10	30	20	200

Day	10	40	0	20	10	100
7						

Ankit had a loss of Rs. 400 on a day in a week.

Hence, Ankit must buy the following number of shares each day:

Day 1: 10 shares

Day 2: 20 shares

Day 3: 30 shares

Day 4: 40 shares

Day 5: 50 shares

Day 6: 60 shares

Day 7: 70 shares

Therefore, the total profit earned by Ankit in the week = $-160 + 400 + 600 - 400 - 1000 + 1200 + 700$

= Rs. 1,340.

Video Solution:**Q6. Text Solution:**

In quadrant I, Sale and Profit both increase from 2021 to 2022

But in case of company X, increase in sales is more than increase in profit. Therefore, company X has definitely expenses increased.

In quadrant II, Profit increases despite of decrease in Sales from 2021 to 2022, so there is a conclusion that Expense decreases definitely.

In quadrant III, Sale and Profit, both decrease from 2021 to 2022

But in case of company R, decrease in profit is more than decrease in sales. Therefore, company R has definitely expenses increased.

In quadrant IV, Profit decreases despite of increase in Sale from 2021 to 2022, so there is a conclusion that Expense increases definitely.

Hence, for companies X, R, S, and U the expenses increase definitely.

Video Solution:

**Q7. Text Solution:**

In quadrant I, Sale and Profit both increase from 2021 to 2022

But in case of company Y and T, increase in profit is more than increase in sales. Therefore, company Y and T has definitely expenses decreased.

In quadrant II, Profit increases despite of decrease in Sale from 2021 to 2022, so there is a conclusion that Expense decreases definitely.

In quadrant III, Sale and Profit, both decrease from 2021 to 2022

But in case of company Z, decrease in sales is more than decrease in profit. Therefore, company Z has definitely expenses decreased.

In quadrant IV, Profit decreases despite of increase in Sale from 2021 to 2022, so there is a conclusion that Expense increases definitely. Hence, for companies Y, T, V, Q, W, and Z the expenses decrease definitely.

Video Solution:**Q8. Text Solution:**

In quadrant I, Sale and Profit both increase from 2021 to 2022

But in case of company X, increase in sales is more than increase in profit. Therefore, company X has definitely expenses increased.

Therefore, expense in 2021 will be more than ₹1000 Crores.

Video Solution:**Q9. Text Solution:**

In quadrant II, Profit increases despite of decrease in Sale from 2021 to 2022, so there is a conclusion that Expense decreases definitely. Company Q is in quadrant II, where there is a definite decrease in expenditure. Therefore, expense of company Q in 2021 will be less than ₹100 Crores

So out of all the options we have only option B which is less than 100 crores.

Video Solution:**Q10. Text Solution:**

In quadrant II, Profit increases despite of decrease in Sale from 2021 to 2022, so there is a conclusion that Expense decreases definitely.

In quadrant IV, Profit decreases despite an increase in Sale from 2021 to 2022, so there is a conclusion that Expense increases definitely.

Since the expenditure of company S (S in quadrant III) increases in the year 2022 and that of Q (Q in II quadrant) decreases in the year 2022.

So, Expenditure of S > Expenditure of Q

Video Solution:

Q11. Text Solution:

Item	Bubble Radius	
	For shopkeeper P	For shopkeeper Q
A	$50 \times \frac{1}{3-1} = 25$	$25 + 50 = 75$
B	$50 \times \frac{2}{2-1} = 100$	$100 - 50 = 50$
C	$75 \times \frac{1}{2-1} = 75$	$75 + 75 = 150$
D	$200 \times \frac{1}{5-1} = 50$	$200 + 50 = 250$
E	$100 \times \frac{2}{2-1} = 200$	$200 - 100 = 100$

For shopkeeper P :

Item	Discount (in Rs.)	Profit (in Rs.)	Marked Price (in Rs.)	Selling Price (in Rs.)	Cost Price (in Rs.)
A	120	60	$3.14 \times 25^2 = 1962.5$	$1962.5 - 120 = 1842.5$	$1842.5 - 60 = 1782.5$
B	80	100	$3.14 \times 100^2 = 31400$	$31400 - 80 = 31320$	$31320 - 100 = 31220$
C	140	150	$3.14 \times 75^2 = 17662.5$	$17662.5 - 140 = 17522.5$	$17522.5 - 150 = 17372.5$
D	200	180	$3.14 \times 50^2 = 7850$	$7850 - 200 = 7650$	$7650 - 180 = 7470$
E	360	250	$3.14 \times 200^2 = 125600$	$125600 - 360 = 125240$	$125240 - 250 = 124990$

For Shopkeeper Q :

Item	Discount (in Rs.)	Profit (in Rs.)	Marker Price (in Rs.)	Selling Price (in Rs.)	Cost Price (in Rs.)
A	120	160	$3.14 \times 75^2 = 17662.5$	$17662.5 - 120 = 17542.5$	$17542.5 - 160 = 17382.5$
B	80	210	$3.14 \times 50^2 = 7850$	$7850 - 80 = 7770$	$7770 - 210 = 7560$
C	140	40	$3.14 \times 150^2 = 70650$	$70650 - 140 = 70510$	$70510 - 40 = 70470$
D	200	80	$3.14 \times 250^2 = 196250$	$196250 - 200 = 196050$	$196050 - 80 = 195970$
E	360	170	$3.14 \times 100^2 = 31400$	$31400 - 360 = 31040$	$31040 - 170 = 30870$

For shopkeeper P:

CP of product C = 17372.5

For shopkeeper Q:

CP of product C = 70470

Required Difference = 70470 - 17372.5 = 53097.5

Video Solution:



Q12. Text Solution:

Item	Bubble Radius	
	For shopkeeper P	For shopkeeper Q
A	$50 \times \frac{1}{3-1} = 25$	$25 + 50 = 75$
B	$50 \times \frac{2}{2-1} = 100$	$100 - 50 = 50$
C	$75 \times \frac{1}{2-1} = 75$	$75 + 75 = 150$
D	$200 \times \frac{1}{5-1} = 50$	$200 + 50 = 250$
E	$100 \times \frac{2}{2-1} = 200$	$200 - 100 = 100$

For shopkeeper P :

Item	Discount (in Rs.)	Profit (in Rs.)	Marked Price (in Rs.)	Selling Price (in Rs.)	Cost Price (in Rs.)
A	120	60	$3.14 \times 25^2 = 1962.5$	$1962.5 - 120 = 1842.5$	$1842.5 - 60 = 1782.5$
B	80	100	$3.14 \times 100^2 = 31400$	$31400 - 80 = 31320$	$31320 - 100 = 31220$
C	140	150	$3.14 \times 75^2 = 17662.5$	$17662.5 - 140 = 17522.5$	$17522.5 - 150 = 17372.5$
D	200	180	$3.14 \times 50^2 = 7850$	$7850 - 200 = 7650$	$7650 - 180 = 7470$
E	360	250	$3.14 \times 200^2 = 125600$	$125600 - 360 = 125240$	$125240 - 250 = 124990$

For Shopkeeper Q :

Item	Discount (in Rs.)	Profit (in Rs.)	Marker Price (in Rs.)	Selling Price (in Rs.)	Cost Price (in Rs.)
A	120	160	$3.14 \times 75^2 = 17662.5$	$17662.5 - 120 = 17542.5$	$17542.5 - 160 = 17382.5$
B	80	210	$3.14 \times 50^2 = 7850$	$7850 - 80 = 7770$	$7770 - 210 = 7560$
C	140	40	$3.14 \times 150^2 = 70650$	$70650 - 140 = 70510$	$70510 - 40 = 70470$
D	200	80	$3.14 \times 250^2 = 196250$	$196250 - 200 = 196050$	$196050 - 80 = 195970$
E	360	170	$3.14 \times 100^2 = 31400$	$31400 - 360 = 31040$	$31040 - 170 = 30870$

Cost price of all given items taken together for shopkeeper P

$$= 1782.5 + 31220 + 17372.5 + 7470 + 124990 = 182835$$

Cost price of all given items taken together for shopkeeper Q

$$= 17382.5 + 7560 + 70470 + 195970 + 30870 = 322252.5$$

$$\text{Difference} = 322252.5 - 182835 = 139417.5$$

Therefore, percentage



$$= \frac{139417.5}{322252.5} \times 100 = 43\% \text{ (approx.)}$$

Video Solution:



Q13. Text Solution:

Item	Bubble Radius	
	For shopkeeper P	For shopkeeper Q
A	$50 \times \frac{1}{3-1} = 25$	$25 + 50 = 75$
B	$50 \times \frac{2}{2-1} = 100$	$100 - 50 = 50$
C	$75 \times \frac{1}{2-1} = 75$	$75 + 75 = 150$
D	$200 \times \frac{1}{5-1} = 50$	$200 + 50 = 250$
E	$100 \times \frac{2}{2-1} = 200$	$200 - 100 = 100$

For shopkeeper P :

Item	Discount (in Rs.)	Profit (in Rs.)	Marked Price (in Rs.)	Selling Price (in Rs.)	Cost Price (in Rs.)
A	120	60	$3.14 \times 25^2 = 1962.5$	$1962.5 - 120 = 1842.5$	$1842.5 - 60 = 1782.5$
B	80	100	$3.14 \times 100^2 = 31400$	$31400 - 80 = 31320$	$31320 - 100 = 31220$
C	140	150	$3.14 \times 75^2 = 17662.5$	$17662.5 - 140 = 17522.5$	$17522.5 - 150 = 17372.5$
D	200	180	$3.14 \times 50^2 = 7850$	$7850 - 200 = 7650$	$7650 - 180 = 7470$
E	360	250	$3.14 \times 200^2 = 125600$	$125600 - 360 = 125240$	$125240 - 250 = 124990$

For Shopkeeper Q :

Item	Discount (in Rs.)	Profit (in Rs.)	Marked Price (in Rs.)	Selling Price (in Rs.)	Cost Price (in Rs.)
A	120	160	$3.14 \times 75^2 = 17662.5$	$17662.5 - 120 = 17542.5$	$17542.5 - 160 = 17382.5$
B	80	210	$3.14 \times 50^2 = 7850$	$7850 - 80 = 7770$	$7770 - 210 = 7560$
C	140	40	$3.14 \times 150^2 = 70650$	$70650 - 140 = 70510$	$70510 - 40 = 70470$
D	200	80	$3.14 \times 250^2 = 196250$	$196250 - 200 = 196050$	$196050 - 80 = 195970$
E	360	170	$3.14 \times 100^2 = 31400$	$31400 - 360 = 31040$	$31040 - 170 = 30870$

For shopkeeper P:

New radius for bubble D = 85% of 50 = 42.5

Then, new marked price of D = $3.14 \times 42.5 \times 42.5$
= 5671.625

And, new selling price of D = $5671.625 - 200 = 5471.625$

For shopkeeper Q:

New radius for bubble D = 60% of 250 = 150

Then, new marked price of D = $3.14 \times 150 \times 150 = 70650$

And, new selling price of D = $70650 - 200 = 70450$

Therefore, required difference = $70450 - 5471.625 = \text{Rs. } 64978 \text{ (approx.)}$

Video Solution:



Q14. Text Solution:

Item	Bubble Radius	
	For shopkeeper P	For shopkeeper Q
A	$50 \times \frac{1}{3-1} = 25$	$25 + 50 = 75$
B	$50 \times \frac{2}{2-1} = 100$	$100 - 50 = 50$
C	$75 \times \frac{1}{2-1} = 75$	$75 + 75 = 150$
D	$200 \times \frac{1}{5-1} = 50$	$200 + 50 = 250$
E	$100 \times \frac{2}{2-1} = 200$	$200 - 100 = 100$

For shopkeeper P :

Item	Discount (in Rs.)	Profit (in Rs.)	Marked Price (in Rs.)	Selling Price (in Rs.)	Cost Price (in Rs.)
A	120	60	$3.14 \times 25^2 = 1962.5$	$1962.5 - 120 = 1842.5$	$1842.5 - 60 = 1782.5$
B	80	100	$3.14 \times 100^2 = 31400$	$31400 - 80 = 31320$	$31320 - 100 = 31220$
C	140	150	$3.14 \times 75^2 = 17662.5$	$17662.5 - 140 = 17522.5$	$17522.5 - 150 = 17372.5$
D	200	180	$3.14 \times 50^2 = 7850$	$7850 - 200 = 7650$	$7650 - 180 = 7470$
E	360	250	$3.14 \times 200^2 = 125600$	$125600 - 360 = 125240$	$125240 - 250 = 124990$

For Shopkeeper Q :



Item	Discount (in Rs.)	Profit (in Rs.)	Marker Price (in Rs.)	Selling Price (in Rs.)	Cost Price (in Rs.)
A	120	160	$3.14 \times 75^2 = 17662.5$	$17662.5 - 120 = 17542.5$	$17542.5 - 160 = 17382.5$
B	80	210	$3.14 \times 50^2 = 7850$	$7850 - 80 = 7770$	$7770 - 210 = 7560$
C	140	40	$3.14 \times 150^2 = 70650$	$70650 - 140 = 70510$	$70510 - 40 = 70470$
D	200	80	$3.14 \times 250^2 = 196250$	$196250 - 200 = 196050$	$196050 - 80 = 195970$
E	360	170	$3.14 \times 100^2 = 31400$	$31400 - 360 = 31040$	$31040 - 170 = 30870$

Average of new selling price of item D for shopkeeper P and Q together

$$= \frac{(7470 - 4500 + 195970 - 5200)}{2} = 96870$$

Video Solution:



Q15. Text Solution:

Item	Bubble Radius	
	For shopkeeper P	For shopkeeper Q
A	$50 \times \frac{1}{3-1} = 25$	$25 + 50 = 75$
B	$50 \times \frac{2}{2-1} = 100$	$100 - 50 = 50$
C	$75 \times \frac{1}{2-1} = 75$	$75 + 75 = 150$
D	$200 \times \frac{1}{5-1} = 50$	$200 + 50 = 250$
E	$100 \times \frac{2}{2-1} = 200$	$200 - 100 = 100$

For shopkeeper P :

Item	Discount (in Rs.)	Profit (in Rs.)	Marked Price (in Rs.)	Selling Price (in Rs.)	Cost Price (in Rs.)
A	120	60	$3.14 \times 25^2 = 1962.5$	$1962.5 - 120 = 1842.5$	$1842.5 - 60 = 1782.5$
B	80	100	$3.14 \times 100^2 = 31400$	$31400 - 80 = 31320$	$31320 - 100 = 31220$
C	140	150	$3.14 \times 75^2 = 17662.5$	$17662.5 - 140 = 17522.5$	$17522.5 - 150 = 17372.5$
D	200	180	$3.14 \times 50^2 = 7850$	$7850 - 200 = 7650$	$7650 - 180 = 7470$
E	360	250	$3.14 \times 200^2 = 125600$	$125600 - 360 = 125240$	$125240 - 250 = 124990$

For Shopkeeper Q :

Item	Discount (in Rs.)	Profit (in Rs.)	Marker Price (in Rs.)	Selling Price (in Rs.)	Cost Price (in Rs.)
A	120	160	$3.14 \times 75^2 = 17662.5$	$17662.5 - 120 = 17542.5$	$17542.5 - 160 = 17382.5$
B	80	210	$3.14 \times 50^2 = 7850$	$7850 - 80 = 7770$	$7770 - 210 = 7560$
C	140	40	$3.14 \times 150^2 = 70650$	$70650 - 140 = 70510$	$70510 - 40 = 70470$
D	200	80	$3.14 \times 250^2 = 196250$	$196250 - 200 = 196050$	$196050 - 80 = 195970$
E	360	170	$3.14 \times 100^2 = 31400$	$31400 - 360 = 31040$	$31040 - 170 = 30870$

The difference of the cost prices of the item B for the shopkeeper P and Q

$$= \text{Rs. } (31220 - 7560)$$

$$= \text{Rs. } 23660$$

Video Solution:



Q16. Text Solution:

Minimum power generated by

$$P = 2MW = 2000KW$$

Remaining power after loss = 75% of
 $2000 = 1500KW$

$$x = \frac{1500}{12}$$

$$x = 125$$

Power generated by Q at the end of experiment
 $= 2MW = 2000KW$

Remaining power after loss = 75% of
 $2000 = 1500KW$

$$Y = \frac{1500}{10}$$

$$Y = 150$$

Required ratio = $x : y = 125 : 150 = 5 : 6$

Video Solution:



**Q17. Text Solution:**

Maximum power generated by

$$R = 11MW = 11000KW$$

Remaining power after loss = 80% of
 $11000 = 8800KW$

According to the question:

$$x = \frac{8800}{88}$$

$$x = 100$$

Now, the number of machines that can be run when each machine starts consuming $12kw$ more power.

$$= \frac{8800}{88 + 12}$$

$$= 88$$

$$\text{Required percent} = \frac{100-88}{100} \times 100 = 12\%$$

Video Solution:**Q18. Text Solution:**

Power generated by S when the experiment starts = $5MW = 5000KW$

Remaining power after loss = 75% of
 $5000 = 3750KW$.

Let the number of type A and type B machines are ' x ' each.

According to the question:

$$12x + 13x = 3750$$

$$x = 150$$

Required number of machines

$$= 2x$$

$$= 300$$

Video Solution:**Q19. Text Solution:**

Power generated by S when generating minimum power = $3MW = 3000KW$

Remaining power after loss = 75% of
 $3000 = 2250KW$

$$x = \frac{2250}{15}$$

$$x = 150$$

Power generated by T when generating maximum power = $15MW = 15000KW$

Remaining power after loss = 80% of
 $15000 = 12000KW$

$$\text{So, } y = \frac{12000}{24}$$

$$\Rightarrow y = 500$$

$$\text{Hence, } x : y = 150 : 500$$

$$= 3 : 10$$

Video Solution:**Q20. Text Solution:**

Maximum power generated by

$$T = 15MW = 15000KW$$

Remaining power after loss = 80% of



$$15000 = 12000KW$$

According to the question:

$$x = \frac{12000}{60}$$

$$x = 200$$

Now, the number of machines that can be run when there is problem in the machines

$$= \frac{12000}{60 + 40}$$

$$= 120$$

$$\text{Required percent} = \frac{200-120}{200} \times 100 = 40\%$$

Video Solution:



Q21. Text Solution:

	Total capacity (in litres)	Amount of electricity (in units) which can be generated per 100 litres of water
A	12000	20
B	45000	15
C	24000	30
D	20000	10

Amount of water in dam 'A' = $0.45 \times 12000 = 5400$

Amount of water used to produce electricity = $\frac{5400}{2} = 2700$ litres

Amount of electricity which can be generated from dam A = $2700 \times \left(\frac{20}{100}\right) = 540$ units

Video Solution:



Q22. Text Solution:

	Total capacity (in litres)	Amount of electricity (in units) which can be generated per 100 litres of water
A	12000	20
B	45000	15
C	24000	30
D	20000	10

Amount of electricity produced from dam 'B' = $45000 \times \frac{15}{100} = 6750$ units

Amount of electricity which can be produced from dam 'D' = $20000 \times \frac{10}{100} = 2000$ units

Required ratio = $6750:2000 = 27 : 8$.

Video Solution:



Q23. Text Solution:

	Total capacity (in litres)	Amount of electricity (in units) which can be generated per 100 litres of water
A	12000	20
B	45000	15
C	24000	30
D	20000	10

Amount of water available in dam 'C' = $24000 - 6000 = 18000$ litres

Amount of electricity which can be generated from available water in dam 'C' = $18000 \times \frac{30}{100} = 5400$ units

Amount earned = $5400 \times 5 = \text{Rs. } 27000$.

Video Solution:





Q24. Text Solution:

Dam	Total capacity (in litres)	Amount of electricity (in units) which can be generated per 100 litres of water
A	12000	20
B	45000	15
C	24000	30
D	20000	10

Required average = $\frac{12000+45000+24000+20000}{4} = 25250$ litres.

Hence, the answer is **none of these**.

Video Solution:



Q25. Text Solution:

Dam	Total capacity (in litres)	Amount of electricity (in units) which can be generated per 100 litres of water
A	12000	20
B	45000	15
C	24000	30
D	20000	10

Amount of electricity which can be produced from dam 'B' = $45000 \times \frac{15}{100} = 6750$ units

Amount of electricity which can be produced from dam 'D' = $20000 \times \frac{10}{100} = 2000$ units

Required difference = **4750** units.

Video Solution:



Q26. Text Solution:

Accuracy % is defined as

$$\frac{\text{Number of Correct Questions}}{\text{Total number of questions attempted}} \times 100$$

By the graph, we can calculate the total number of attempts by each member.

For Group A member 1: - Accuracy = 25%,

Number of correct answers = 2

Total attempts = 8, Number of incorrect questions = $8 - 2 = 6$

For Group A member 2: - Accuracy = 66.66%,

Number of correct questions = 6

Total attempts = 9, Number of incorrect questions = $9 - 6 = 3$.

Similarly, we can calculate for each member of all the groups.

Let us summarise the result in the following table-

Member	Attempt	Correct	Incorrect	Un-attempted
Group A member 1	8	2	6	2
Group A member 2	9	6	3	1
Group A member 3	10	10	0	0
Group B member 1	8	1	7	2
Group B member 2	5	5	0	5
Group B member 3	10	9	1	0
Group C member 1	6	3	3	4
Group C member 2	8	5	3	2
Group C member 3	8	6	2	2

Accuracy for Group

$$A = \frac{2+6+10}{8+9+10} \times 100 = 66.66\%$$

Accuracy for Group B

$$= \frac{1+5+9}{8+5+10} \times 100 = 65.22\%$$

Accuracy for Group C

$$= \frac{3+5+6}{6+8+8} \times 100 = 63.64\%$$

Accuracy of Group A is highest. So, group A is the winner.

Video Solution:



**Q27. Text Solution:**

Accuracy % is defined as

$$\frac{\text{Number of Correct Questions}}{\text{Total number of questions attempted}} \times 100$$

By the graph, we can calculate the total number of attempts by each member.

For Group A member 1: - Accuracy = 25%,

Number of correct answers = 2

Total attempts = 8, Number of incorrect questions = 8 - 2 = 6

For Group A member 2: - Accuracy = 66.66%,

Number of correct questions = 6

Total attempts = 9, Number of incorrect questions = 9 - 6 = 3.

Similarly, we can calculate for each member of all the groups.

Let us summarise the result in the following table-

Member	Attempt	Correct	Incorrect	Un-attempted
Group A member 1	8	2	6	2
Group A member 2	9	6	3	1
Group A member 3	10	10	0	0
Group B member 1	8	1	7	2
Group B member 2	5	5	0	5
Group B member 3	10	9	1	0
Group C member 1	6	3	3	4
Group C member 2	8	5	3	2
Group C member 3	8	6	2	2

By the above table

Total attempts of Group A = 8+9+10 = 27

Total attempts of Group B = 8+5+10 = 23

Total attempts of Group C = 6+8+8 = 22.

Hence maximum attempt is made by Group A.

Video Solution:**Q28. Text Solution:**

As of now, accuracy of Group A = 66.66% and that of Group C = 63.64%

After swapping, to keep the accuracy of Group C higher than Group A, the member who moved out from Group C should have lower accuracy than the member who joined Group C.

Let us consider the following possibilities-

Possibility {i}

The member with 100% accuracy from Group A is swapped with the member with 50% accuracy from Group C.

New accuracy of Group

$$A = \frac{2+6+3}{8+9+6} \times 100 = 47.83\%$$

New accuracy of Group C

$$= \frac{10+5+6}{10+8+8} \times 100 = 80.76\%$$

Hence it is one of the possible swapping.

Possibility {ii}

The member with 100% accuracy from Group A is swapped with the member with 62.5% accuracy from Group C.

New accuracy of Group

$$A = \frac{2+6+5}{8+9+8} \times 100 = 52\%$$

New accuracy of Group

$$C = \frac{3+10+6}{6+10+8} \times 100 = 79.16\%$$

Hence it is also one of the possible swapping.

Therefore, more than one member can be swapped to make this happen.

Video Solution:**Q29. Text Solution:**

Accuracy % is defined as

$$\frac{\text{Number of Correct Questions}}{\text{Total number of questions attempted}} \times 100$$

By the graph, we can calculate the total number of attempts by each member.

For Group A member 1: - Accuracy = 25%,

Number of correct answers = 2



Total attempts = 8, Number of incorrect questions = $8 - 2 = 6$

For Group A member 2: - Accuracy = 66.66%,
Number of correct questions = 6

Total attempts = 9, Number of incorrect questions = $9 - 6 = 3$.

Similarly, we can calculate for each member of all the groups.

Let us summarise the result in the following table-

Member	Attempt	Correct	Incorrect	Un-attempted
Group A member 1	8	2	6	2
Group A member 2	9	6	3	1
Group A member 3	10	10	0	0
Group B member 1	8	1	7	2
Group B member 2	5	5	0	5
Group B member 3	10	9	1	0
Group C member 1	6	3	3	4
Group C member 2	8	5	3	2
Group C member 3	8	6	2	2

Total correct answers of Group B = $1+5+9 = 15$

Total incorrect answers of Group B = $7+0+1 = 8$

Hence, total score of Group B = $15(4) + 8(-1) = 52$.

Video Solution:



Q30. Text Solution:

Accuracy % is defined as

$$\frac{\text{Number of Correct Questions}}{\text{Total number of questions attempted}} \times 100$$

By the graph, we can calculate the total number of attempts by each member.

For Group A member 1: - Accuracy = 25%,

Number of correct answers = 2

Total attempts = 8, Number of incorrect questions = $8 - 2 = 6$

For Group A member 2: - Accuracy = 66.66%,
Number of correct questions = 6

Total attempts = 9, Number of incorrect questions = $9 - 6 = 3$.

Similarly, we can calculate for each member of all the groups.

Let us summarise the result in the following table-

Member	Attempt	Correct	Incorrect	Un-attempted
Group A member 1	8	2	6	2
Group A member 2	9	6	3	1
Group A member 3	10	10	0	0
Group B member 1	8	1	7	2
Group B member 2	5	5	0	5
Group B member 3	10	9	1	0
Group C member 1	6	3	3	4
Group C member 2	8	5	3	2
Group C member 3	8	6	2	2

Accuracy for Group A = 66.66%

Accuracy for Group B = 65.21%

If 1 more question is correct out of attempted questions by the worst performer of Group B:

$$\text{Accuracy for Group B} = \frac{2+5+9}{8+5+10} \times 100 = 69.57\%$$

Hence at least 1 more question should be answered correctly.

Video Solution:

